GENOMEDL CLINICAL GENOMICS PIPELINE (v3)

Pediatric & Germline Hereditary Cancer Screening Report

Sample Identifier:	HG002-GIAB-BENCHMARK	**Reference Genome:**	GRCh38 / hg38
Panel Profile:	130 Hereditary Cancer Genes	**Analysis Date:**	November 07, 2026 (18:39 UTC)
Total Variants Filtered:	1392 candidates	**Actionable Tiers:**	**0 Pathogenic / Likely Pathogenic**

NO HIGH-PENETRANCE PATHOGENIC VARIANTS IDENTIFIED

No pathogenic or likely pathogenic variants were detected in the 130 evaluated cancer predisposition genes.

2. Detailed Genomic Variant Classifications

Each candidate variant was filtered against gnomAD population frequencies (AF > 1% = Benign), matched against ClinVar curated submissions (≥2 stars), and unclassified novel variants were scored with the GenomeDL v3 ESM-augmented Multimodal Fusion ResNet.

GENE & LOCUS	HGVS / PROTEIN	GT (DP)	CLASSIFICATION	EVIDENCE SOURCE	MODEL P
NTHL1 16:2035679 C>T	Calibrated probability P(pathogenic) = 0	0/1 (Het)	**Carrier (Recessive)**	GenomeDL v3 ESM-Augmented Fusion ResNet	0.030
NF2 22:29603247 G>C	Calibrated probability P(pathogenic) = 0	0/1 (Het)	**VUS**	GenomeDL v3 ESM-Augmented Fusion ResNet	0.421
SDHB 1:17023648 G>A	Curated clinical consensus from ClinVar submission (Variation ID 695745)	0/1 (Het)	**Benign**	ClinVar lookup (2 stars: criteria provided, multiple submitters, no conflicts)	0.001
SDHB 1:17023743 A>G	Calibrated probability P(pathogenic) = 0	0/1 (Het)	**Benign**	GenomeDL v3 ESM-Augmented Fusion ResNet	0.004
SDHB 1:17023748 T>C	Calibrated probability P(pathogenic) = 0	0/1 (Het)	**Benign**	GenomeDL v3 ESM-Augmented Fusion ResNet	0.014
SDHB 1:17027493 T>G	Calibrated probability P(pathogenic) = 0	0/1 (Het)	**Benign**	GenomeDL v3 ESM-Augmented Fusion ResNet	0.004
SDHB 1:17028892 T>C	Calibrated probability P(pathogenic) = 0	0/1 (Het)	**Benign**	GenomeDL v3 ESM-Augmented Fusion ResNet	0.014
SDHB 1:17032891 T>C	Curated clinical consensus from ClinVar submission (Variation ID 1234802)	0/1 (Het)	**Benign**	ClinVar lookup (2 stars: criteria provided, multiple submitters, no conflicts)	0.001
SDHB 1:17033181 C>A	Curated clinical consensus from ClinVar submission (Variation ID 258889)	0/1 (Het)	**Benign**	ClinVar lookup (2 stars: criteria provided, multiple submitters, no conflicts)	0.001
SDHB 1:17044657 T>C	Curated clinical consensus from ClinVar submission (Variation ID 1252797)	0/1 (Het)	**Benign**	ClinVar lookup (2 stars: criteria provided, multiple submitters, no conflicts)	0.001
SDHB 1:17044726 C>T	Curated clinical consensus from ClinVar submission (Variation ID 811493)	0/1 (Het)	**Benign**	ClinVar lookup (2 stars: criteria provided, multiple submitters, no conflicts)	0.001
SDHB 1:17044999 C>T	Curated clinical consensus from ClinVar submission (Variation ID 1279662)	0/1 (Het)	**Benign**	ClinVar lookup (2 stars: criteria provided, multiple submitters, no conflicts)	0.001

**Note: 1,380 additional benign polymorphisms are omitted from this summary table for clinical conciseness. Full records are archived in the structured JSON report.*

4. Computational Methodology & Validation Standards

Model Architecture: Multimodal Fusion ResNet combining 26 tabular biochemical features, syndrome embeddings, and ESM-2 (t12_35M_UR50D) masked-marginal log-likelihood ratios and hidden states.
Calibration & Benchmark: Validation-fitted Temperature Scaling (T=0.9357, ECE=0.0249). On 5-fold Gene-Held-Out cross-validation (GroupKFold), model achieves ROC-AUC **0.9042 [95% CI: 0.8872 – 0.9168]** on unseen genes vs tree baseline 0.7225.

MANDATORY CLINICAL GENOMICS DISCLAIMER:
Research prototype, not a diagnosis. Confirm in a certified clinical diagnostic laboratory with formal genetic counselling. Testing children or minors for adult-onset conditions requires specialized pediatric genetic counselling and informed consent.